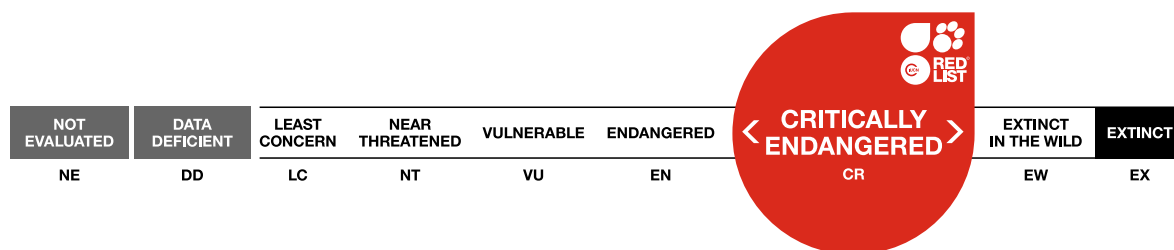


## *Bradypus pygmaeus*, Pygmy Three-toed Sloth

Assessment by: Smith, D., Voirin, B., Chiarello, A. & Moraes-Barros, N.



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## Taxonomy

| Kingdom  | Phylum   | Class    | Order  | Family       |
|----------|----------|----------|--------|--------------|
| Animalia | Chordata | Mammalia | Pilosa | Bradypodidae |

**Scientific Name:** *Bradypus pygmaeus* Anderson & Handley, 2001

### Common Name(s):

- English: Pygmy Three-toed Sloth, Pygmy Three-fingered Sloth
- Spanish; Castilian: Perezoso Pigmeo, Perezoso Pigmeo de Tres Dedos
- German: Zwergfaultier
- Portuguese: Preguiça Anã

### Taxonomic Notes:

*Bradypus pygmaeus* has only recently been described by Anderson and Handley (2001) as a separate species.

## Assessment Information

**Red List Category & Criteria:** Critically Endangered B1ab(ii,iii)+2ab(ii,iii) [ver 3.1](#)

**Year Published:** 2022

**Date Assessed:** April 22, 2022

### Justification:

*Bradypus pygmaeus* is listed as Critically Endangered under criterion B (CR B1ab(ii,iii)+2ab(ii,iii)) as this species has a very restricted range, being found only on one very small island less than 5 km<sup>2</sup> in size, and there is likely a continuing decline in the quality of habitat and area of occupancy due to habitat degradation.

### Previously Published Red List Assessments

[2014 – Critically Endangered \(CR\)](#)

2011 – Critically Endangered (CR)

2008 – Critically Endangered (CR)

2006 – Critically Endangered (CR)

## Geographic Range

### Range Description:

*Bradypus pygmaeus* is known only from Isla Escudo de Veraguas, in the islands of Bocas del Toro, Panama. Sloths on the younger islands of the Bocas del Toro archipelago are conspecific with *Bradypus variegatus*. Isla Escudo de Veraguas has a total area of approximately 4.3 km<sup>2</sup> and is about 17.6 km from the mainland of Panama. For the Red List assessment, the parameters area of occupancy (AOO) and extent of occurrence (EOO) are larger than the total island area because of the methods used to

measure these (IUCN Standards & Petitions Committee 2022). Its AOO is 8 km<sup>2</sup> (total area within two occupied 2x2 km grid cells); and its EOO is 5 km<sup>2</sup> (based on a minimum convex polygon around the entire island), however this is raised to 8 km<sup>2</sup> because EOO should not be smaller than the AOO (IUCN Standards & Petitions Committee 2022).

**Country Occurrence:**

**Native, Extant (resident):** Panama

# Distribution Map

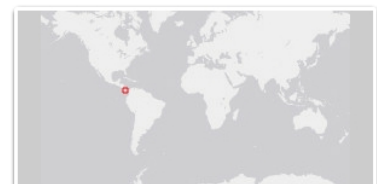
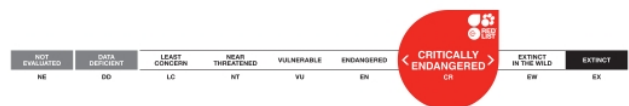


## Legend

■ EXTANT (RESIDENT)

Compiled by:

IUCN (International Union for Conservation of Nature) 2022



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

## Population

There is no accurate information available on the total population size of *Bradypus pygmaeus*. Previous censuses only searched for sloths in mangroves, which comprise less than 3% of the island. Censusing cryptic, slow moving arboreal mammals in dense forest is challenging and limited; however, *B. pygmaeus* has been observed in the interior forests of the island. The population size is most likely between 2,000 and 2,500 individuals (D. Smith pers. comm. 2022). In any case, it is extremely small for an entire species (Voirin *et al.* 2015).

**Current Population Trend:** Decreasing

## Habitat and Ecology (see Appendix for additional information)

This is the smallest of all sloths, and is found both in mangrove patches and on the interior of Isla Escudo. Although previously thought to exclusively inhabit the red mangroves of the island, recent tracking studies have found the sloths on the interior of the island, in dense tropical rainforest. Because of the difficulty of censusing cryptic canopy mammals, their density and abundance in the thicker forests is unknown. Nothing is known about their reproduction, lifespan, home range, or diet, although it is suspected that it primarily, if not exclusively, feeds on mangrove leaves. Its population is likely larger than previously estimated, but is still limited due to its restricted habitat range.

**Systems:** Terrestrial

## Use and Trade (see Appendix for additional information)

The species is not utilized. Although it was previously thought that locals would eat pygmy sloths, this was recently disproven.

## Threats (see Appendix for additional information)

Although the island is uninhabited, there are seasonal visitors (fishermen, campesinos, lobster divers, tourists, and local people) who harvest timber to maintain wooden houses on the island. Preliminary studies suggest a reduced level of genetic diversity for pygmy sloths compared to its putative population of origin, the common sloth population from mainland Panama. This is expected, considering the history of species diversification and isolation on the island. However, signs of a more recent population bottleneck were also detected (Silva *et al.* 2018, N. Moraes-Barros pers. comm. 2013). These results highlight the need of a continuous evaluation of the population status, trends, and additional studies considering a possible scenario of endogamic depression if the (already low) population size decreases any further.

Despite having been designated as a protected landscape through a governmental resolution in 2009, a number of domestic and international efforts have been mounted to develop tourism infrastructure on the island. The increased interest in using the island for touristic purposes exponentially increases the number of local visitors and tourists. On the other hand, the intentions of local indigenous people to build permanent settlements on the island have increased the number of permitted constructions, putting the pygmy sloths and their habitat at risk (D. Smith pers. comm. 2022). The current status of the island's custody is vague; a governmental resolution, and thus the protected status of the island, cannot be revoked, but no government staff has been appointed specifically to enforce protection of the island.

Ongoing disagreements between the local Ngäbe bugle Comarca, regional politicians, and the Panamanian government are further complicating long-term protection of the island and the pygmy sloths. Additionally, as pygmy sloths have become more widely recognized internationally, there is growing interest in collecting them for captivity.

## **Conservation Actions (see Appendix for additional information)**

*Bradypus pygmaeus* is endemic to a single island of Panama, which is protected as a wildlife refuge and is contained within the Ngäbe bugle Comarca. There is a need to improve the enforcement of this protected area. A comprehensive conservation plan is underway, bringing together the local community, wildlife authorities in Panama, and the national and international scientific community to protect the island, using the pygmy sloth as a flagship species.

The pygmy sloth is listed on CITES Appendix II (Notification to the Parties 2013/052, 20 November 2013).

## **Credits**

|                                        |                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Assessor(s):</b>                    | Smith, D., Voirin, B., Chiarello, A. & Moraes-Barros, N.                                                                                                                                                              |
| <b>Reviewer(s):</b>                    | Abba, A.M. & Morales-Campos, J.                                                                                                                                                                                       |
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| <b>Facilitator(s) and Compiler(s):</b> | Tognelli, M.F. & Superina, M.                                                                                                                                                                                         |
| <b>Partner(s) and Institution(s):</b>  | IUCN SSC Anteater, Sloth and Armadillo Specialist Group                                                                                                                                                               |
| <b>Authority/Authorities:</b>          | IUCN SSC Anteater, Sloth and Armadillo Specialist Group                                                                                                                                                               |

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## External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

## Appendix

### Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Habitat                                                                                   | Season   | Suitability | Major Importance? |
|-------------------------------------------------------------------------------------------|----------|-------------|-------------------|
| 1. Forest -> 1.6. Forest - Subtropical/Tropical Moist Lowland                             | Resident | Unknown     | -                 |
| 1. Forest -> 1.7. Forest - Subtropical/Tropical Mangrove Vegetation Above High Tide Level | Resident | Suitable    | Yes               |

### Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Threat                                                                                                                      | Timing    | Scope                                                                                                     | Severity            |
|-----------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------|---------------------|
| 1. Residential & commercial development -> 1.3. Tourism & recreation areas                                                  | Ongoing   | Whole (>90%)                                                                                              | Unknown             |
|                                                                                                                             | Stresses: | 1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance    |                     |
| 5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.1. Intentional use (species is the target) | Ongoing   | Minority (<50%)                                                                                           | Negligible declines |
|                                                                                                                             | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                             |                     |
| 5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.5. Motivation Unknown/Unrecorded                        | Ongoing   | Unknown                                                                                                   | Unknown             |
|                                                                                                                             | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation |                     |

### Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Conservation Action in Place                          |
|-------------------------------------------------------|
| In-place research and monitoring                      |
| Systematic monitoring scheme: Yes                     |
| In-place land/water protection                        |
| Conservation sites identified: Yes, over entire range |
| Occurs in at least one protected area: Yes            |
| In-place species management                           |
| Harvest management plan: No                           |
| Successfully reintroduced or introduced benignly: No  |

|                                                               |
|---------------------------------------------------------------|
| <b>Conservation Action in Place</b>                           |
| Subject to ex-situ conservation: No                           |
| In-place education                                            |
| Subject to any international management / trade controls: Yes |

## Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Conservation Action Needed                                                      | Notes |
|---------------------------------------------------------------------------------|-------|
| 2. Land/water management -> 2.1. Site/area management                           | -     |
| 3. Species management -> 3.1. Species management -> 3.1.1. Harvest management   | -     |
| 4. Education & awareness -> 4.3. Awareness & communications                     | -     |
| 5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.2. National level     | -     |
| 5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.3. Sub-national level | -     |

## Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Research Needed                                               | Notes |
|---------------------------------------------------------------|-------|
| 1. Research -> 1.1. Taxonomy                                  | -     |
| 1. Research -> 1.2. Population size, distribution & trends    | -     |
| 1. Research -> 1.3. Life history & ecology                    | -     |
| 1. Research -> 1.4. Harvest, use & livelihoods                | -     |
| 1. Research -> 1.5. Threats                                   | -     |
| 1. Research -> 1.6. Actions                                   | -     |
| 2. Conservation Planning -> 2.1. Species Action/Recovery Plan | -     |
| 2. Conservation Planning -> 2.2. Area-based Management Plan   | -     |
| 3. Monitoring -> 3.1. Population trends                       | -     |
| 3. Monitoring -> 3.3. Trade trends                            | -     |
| 3. Monitoring -> 3.4. Habitat trends                          | -     |

## Additional Data Fields

|                                                                   |
|-------------------------------------------------------------------|
| <b>Distribution</b>                                               |
| Estimated area of occupancy (AOO) (km <sup>2</sup> ): 8           |
| Continuing decline in area of occupancy (AOO): Yes                |
| Extreme fluctuations in area of occupancy (AOO): No               |
| Estimated extent of occurrence (EOO) (km <sup>2</sup> ): 8        |
| Continuing decline in extent of occurrence (EOO): No              |
| Extreme fluctuations in extent of occurrence (EOO): No            |
| Number of Locations: 1                                            |
| Continuing decline in number of locations: No                     |
| Extreme fluctuations in the number of locations: No               |
| Lower elevation limit (m): 0                                      |
| <b>Population</b>                                                 |
| Extreme fluctuations: No                                          |
| Population severely fragmented: No                                |
| All individuals in one subpopulation: Yes                         |
| <b>Habitats and Ecology</b>                                       |
| Continuing decline in area, extent and/or quality of habitat: Yes |

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